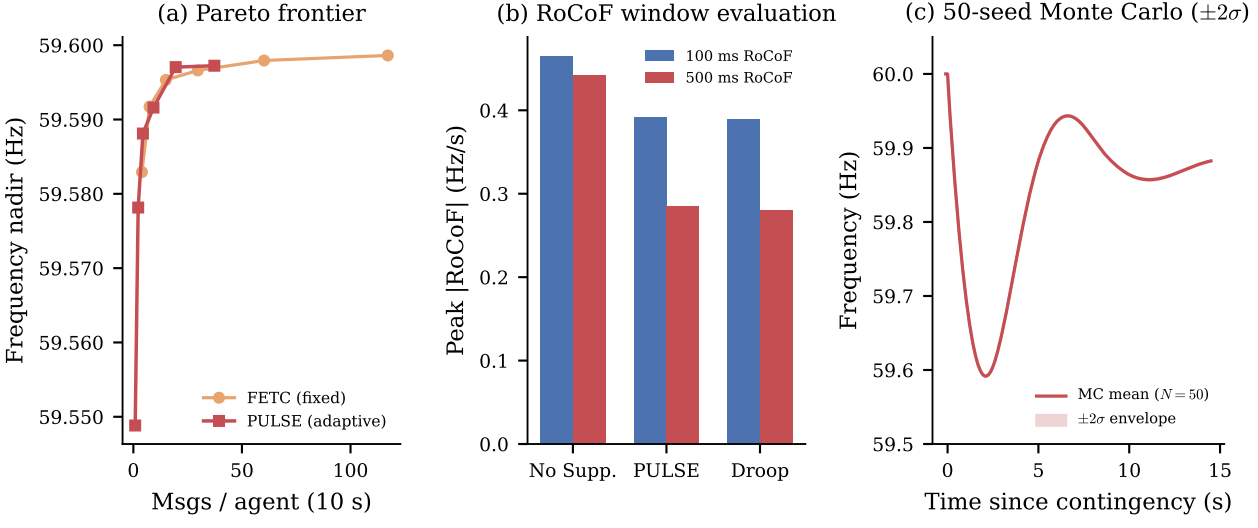

(a) Pareto frontier
Frequency nadir (Hz)
59.600
59.590
59.580
59.570
59.560
59.550
0
50
100
Msgs / agent (10 s)
FETC (fixed)
PULSE (adaptive)
(b) RoCoF window evaluation
Peak |RoCoF| (Hz/s)
0.4
0.3
0.2
0.1
0.0
100 ms RoCoF
500 ms RoCoF
No Supp.
PULSE
Droop
(c) 50-seed Monte Carlo (±2σ)
Frequency (Hz)
60.0
59.9
59.8
59.7
59.6
59.5
0
5
10
15
Time since contingency (s)
MC mean (N = 50)
±2σ envelope